40-question answer key and explanation guide

REVIEW DRAFT. Source-authored explanations retained; independent academic validation is pending.

01-D 02-C 03-B 04-A 05-B 06-D 07-A 08-C
09-D 10-A 11-D 12-C 13-A 14-C 15-C 16-C
17-D 18-D 19-A 20-A 21-D 22-A 23-B 24-A
25-D 26-D 27-B 28-B 29-A 30-B 31-B 32-B
33-C 34-A 35-D 36-C 37-B 38-C 39-C 40-B

1. A severe rise in intracranial pressure produces systemic hypertension with reflex bradycardia. What is this response?

Answer: D. Cushing reflex.

Explanation: The Cushing response reflects brainstem ischaemia driving systemic hypertension, with baroreflex-mediated bradycardia. Brain perfusion is threatened as raised intracranial pressure compresses cerebral vessels.

Not A: Pulmonary hypoxic vasoconstriction is a local lung response.

Not B: Atrial stretch does not explain raised ICP-driven hypertension.

Not C: Bainbridge responds to increased atrial filling.

Source: lecture-6:Q17 | 22-23

2. Which pressure relationship helps prevent fluid accumulation in normal pulmonary alveoli?

Answer: C. Relatively low pulmonary capillary hydrostatic pressure.

Explanation: Low pulmonary capillary hydrostatic pressure relative to plasma oncotic forces helps minimise filtration into air spaces.

Not A: Pulmonary capillary pressure is much lower than systemic arterial pressure.

Not B: Plasma proteins contribute the inward oncotic force.

Not D: Alveolar spaces normally contain little free fluid and protein.

Source: lecture-6:Q32 | 31

3. A cuff that is far too small is used on a large upper arm. Which measurement error is likely?

Answer: B. An undersized cuff gives a falsely high reading.

Explanation: An undersized cuff transmits pressure poorly around a large arm and commonly overestimates measured pressure.

Not A: Cuff size materially affects readings.

Not C: Both systolic and diastolic measurements may be distorted.

Not D: A small cuff usually overestimates rather than underestimates BP.

Source: circulation-1:Q64 | 79-80; Chapter 10 ?8

4. The CSF volume is about 150 mL and production is 550 mL/day. Approximately how many complete turnovers occur per day?

Answer: A. About 4.

Explanation: Dividing daily production by total CSF volume gives 550/150 ~ 3.7 turnovers each day.

Not B: Two turnovers would require about 300 mL/day.

Not C: One turnover would represent only about 150 mL/day.

Not D: Six turnovers would correspond to around 900 mL/day.

Source: lecture-6:Q8 | 14

5. Increasing capillary hydrostatic pressure while other Starling forces stay unchanged tends to produce what?

Answer: B. More fluid filtration into tissues.

Explanation: A higher capillary hydrostatic force favours filtration across the microvascular wall toward tissue. Raising venous pressure is one way to produce this hydrostatic change.

Not A: The pressure favours outward, not reduced, filtration.

Not C: Lymphatic flow generally responds to excess tissue fluid rather than reversing.

Not D: Plasma oncotic pressure does not automatically increase.

Source: circulation-1:Q82 | 85-88

6. A patient develops cardiac tamponade. Why can stroke volume fall despite a normal blood volume?

Answer: D. High pericardial pressure restricts ventricular filling.

Explanation: Elevated intrapericardial pressure limits end-diastolic filling and preload, reducing stroke volume. The restricted ventricular volume prevents an adequate Frank-Starling response.

Not A: Tamponade reduces the ability of ventricles to expand.

Not B: Venous return and effective filling become impaired.

Not C: Valvular opening depends on pressure, but reduced preload is the central problem.

Source: circulation-1:Q38 | 43-44; Chapter 7 ?6

7. Arterial pressure inside an arteriole is unchanged, but pressure in its surrounding tissue rises. What happens to transmural pressure?

Answer: A. It decreases because the pressure difference narrows.

Explanation: Transmural pressure equals internal pressure minus surrounding tissue pressure; increasing the latter reduces the difference.

Not B: A higher external pressure reduces, rather than widens, the difference.

Not C: It equals internal pressure only when external pressure is zero.

Not D: Transmural pressure depends on both internal and external pressure.

Source: circulation-2:Q6 | 8-10

8. A severely underperfused patient begins breathing rapidly, although the lungs were initially healthy. What drives this change?

Answer: C. Lactic acidosis stimulates peripheral chemoreceptors.

Explanation: Inadequate tissue oxygen delivery promotes anaerobic metabolism and lactic acidosis. The resulting acidemia can stimulate peripheral chemoreceptors and increase ventilation; reduced oxygen delivery alone is not equivalent to low arterial oxygen tension.

Not A: Aerobic metabolism is limited by inadequate oxygen delivery.

Not B: Chemoreceptor stimulation, rather than suppression, promotes respiration.

Not D: Excess oxygen is not the initiating problem in shock.

Source: circulation-3:Q25 | Circulation 3, slides 23

9. How does local hypoxia affect most systemic arterioles versus pulmonary arterioles?

Answer: D. Systemic vessels dilate; pulmonary vessels constrict.

Explanation: Systemic hypoxic tissues commonly dilate to improve supply, whereas low alveolar oxygen constricts adjacent pulmonary arterioles to improve ventilation-perfusion matching.

Not A: The response directions have been reversed.

Not B: Systemic tissues usually need increased flow in hypoxia.

Not C: Oxygen tension influences both beds, but their dominant local responses differ.

Source: lecture-6:Q37 | 32; Chapter 24 ?5

10. A pressure tracing shows atrial contraction shortly after an ECG P wave and ventricular contraction shortly after QRS. What principle explains both delays?

Answer: A. Electrical excitation normally precedes the mechanical response of cardiac muscle.

Explanation: The electrical depolarisation that activates cardiac muscle precedes its force development. P precedes atrial systole, and QRS precedes ventricular systole.

Not B: This reverses the normal electrical-to-mechanical sequence.

Not C: The T wave reflects ventricular electrical recovery, after ventricular depolarisation.

Not D: Pressure changes follow electrical activation and are not what skin ECG leads measure.

Source: lecture-7:Q34 | 6-11; Chapter 6 ?5

11. What extra perspective do the chest leads V1-V6 add to the standard limb-lead views?

Answer: D. They show cardiac electrical vectors in the horizontal plane.

Explanation: The limb leads provide frontal-plane views, while the precordial leads examine the same heart from different points around the horizontal plane.

Not A: Chest leads, like limb leads, still record surface electrical potentials.

Not B: Frontal and horizontal views complement rather than replace one another.

Not C: Both sets of leads record combined cardiac electrical events.

Source: lecture-7:Q10 | 3-5; Chapter 6 ?3

12. During progressive shock, capillary hydrostatic pressure paradoxically rises while systemic arterial pressure is falling. Which microvascular pattern best explains it?

Answer: C. Open capillary entry with constricted venules.

Explanation: With capillary entry relatively open but venular outflow restricted, blood stagnates and local capillary pressure rises, driving fluid into the interstitium.

Not A: Lower entry with free outflow would not cause the described stasis.

Not B: Greater drainage would reduce rather than increase local stagnation.

Not D: Both reduced entry and improved drainage favour lower capillary pressure.

Source: circulation-3:Q37 | Circulation 3, slide 26; Chapter 20 ?5

13. Lead II has an upright QRS while aVR has a downward QRS during the same beat. What best explains the difference?

Answer: A. Each lead projects the same electrical vector onto a different axis.

Explanation: A lead is a directional electrical view. The normal mean ventricular depolarisation vector points generally towards Lead II and away from aVR.

Not B: Both QRS complexes represent ventricular depolarisation, despite opposite polarity.

Not C: Both leads sample the electrical activity of the same heartbeat.

Not D: Lead orientation does not change the sequence of atrial and ventricular activation.

Source: lecture-7:Q4 | 3-5; Chapter 6 ?2

14. A patient with marked polycythaemia has elevated haematocrit. What is the expected direct effect on systemic flow resistance?

Answer: C. Higher viscosity and higher resistance.

Explanation: More red cells increase blood viscosity and consequently vascular resistance, raising the heart's workload.

Not A: Both properties generally increase, not decrease.

Not B: Higher viscosity raises resistance rather than lowering it.

Not D: Lower viscosity does not follow from higher haematocrit.

Source: circulation-1:Q56 | 71-72

15. A resting muscle suddenly becomes active while systemic arterial pressure stays steady. What most directly increases its blood supply?

Answer: C. Its local arterioles reduce their resistance.

Explanation: Flow = pressure difference / resistance. With arterial pressure unchanged, the only way to get more flow is lower resistance, and the resistance that tissues control is in their arterioles. Metabolites from the working muscle relax those arterioles.

Not A: Vein compliance mainly affects stored volume, not the main resistance to flow.

Not B: The stem states that systemic arterial pressure remains steady.

Not D: Capillaries lack the smooth muscle needed for this response.

Source: circulation-2:Q1 | 1-4

16. At one instant, Lead I records +0.4 mV and Lead III records -0.2 mV. What must Lead II record?

Answer: C. +0.2 mV, from adding the signed voltages.

Explanation: Einthoven's law is V(II) = V(I) + V(III). Therefore +0.4 + (-0.2) = +0.2 mV at the same instant.

Not A: Lead II is obtained by combining the two measured limb-lead voltages.

Not B: The negative sign of Lead III must be retained in the sum.

Not D: This reverses the sign of the result rather than adding the signed voltages.

Source: lecture-7:Q8 | 3; Chapter 6 ?3

17. A region of heart muscle starts working harder. Which local product of its own metabolism helps increase its blood flow?

Answer: D. Adenosine formed as ATP is used up.

Explanation: Working muscle uses ATP faster, and its breakdown products (adenosine, ADP, AMP) build up and dilate the local arterioles. Flow rises in proportion to how hard the tissue works, which is why the heart relies so heavily on this.

Not A: Platelet serotonin constricts vessels, and it comes from platelets, not from the working muscle.

Not B: Angiotensin II is a circulating constrictor, not a product of the muscle's metabolism.

Not C: Noradrenaline is a nerve transmitter that mainly constricts arterioles via alpha1 receptors.

Source: circulation-2:Q9 | 6

18. What physiological purpose is served by slow conduction through the AV node?

Answer: D. It allows atrial contraction to finish ventricular filling.

Explanation: The AV delay separates atrial and ventricular activation so atrial systole can augment ventricular filling before ventricular systole.

Not A: Normal impulses subsequently travel through His and Purkinje fibres.

Not B: The sequence is atrial activation before ventricular activation.

Not C: The aortic valve remains closed during ordinary atrial systole.

Source: circulation-1:Q14 | 8-10; Chapter 3 ?5

19. Which statement best reflects the updated interpretation of capillary fluid balance?

Answer: A. The glycocalyx regulates filtration; lymph returns excess fluid.

Explanation: The revised Starling model includes the glycocalyx and emphasises that sustained net filtration is normally cleared by lymphatics.

Not B: Lymphatic return remains vital even at normal pressures.

Not C: The classical arterial filtration-venous reabsorption picture overstates sustained reabsorption.

Not D: Protein distributions can influence effective oncotic gradients.

Source: circulation-1:Q84 | 85-88; Chapter 13 ?6 (modern clarification)

20. Minutes after a severe allergic exposure, a patient's pressure drops with generalised swelling. Which vascular action is most relevant?

Answer: A. Histamine-related dilatation with increased capillary permeability.

Explanation: Anaphylaxis releases mediators including histamine that widen vessels and increase capillary leakage, reducing effective perfusion.

Not B: Anaphylaxis typically causes dilation and increased leak.

Not C: Higher pump performance would not explain the sudden widespread leakage.

Not D: Occult bleeding is not the primary allergic vascular mechanism.

Source: circulation-3:Q46 | Circulation 3, slides 29

21. How does normal inspiration tend to assist systemic venous return?

Answer: D. Lower chest pressure draws venous blood toward the heart.

Explanation: Inspiration decreases intrathoracic pressure and promotes a pressure gradient drawing blood from peripheral veins to the thorax.

Not A: Lower rather than higher thoracic pressure assists inflow.

Not B: Venous valves do not all close in inspiration.

Not C: Return is generally supported rather than abolished.

Source: circulation-1:Q76 | 56, 97-98

22. An exercising leg receives increased blood flow while its muscle continues contracting. What is this response called?

Answer: A. Active hyperaemia driven by ongoing tissue metabolism.

Explanation: Active hyperaemia is the rise in flow while a tissue is working: its metabolites dilate the arterioles for as long as activity continues. Reactive hyperaemia is the surge after a blockage is released.

Not B: Reactive hyperaemia is a surge after an obstruction ends.

Not C: The increase originates largely from local demand, not a pressure-rise baroreflex.

Not D: The flow rises above baseline rather than staying constant.

Source: circulation-2:Q13 | 11

23. Why does persistent lactic acidosis make circulatory shock harder to reverse?

Answer: B. It depresses contraction and catecholamine responses.

Explanation: Acidosis from anaerobic metabolism weakens cardiac pumping and reduces vascular response to catecholamines, undermining compensation.

Not A: Acidosis does not reliably prevent fluid movement into tissues.

Not C: Aerobic ATP generation still requires oxygen.

Not D: Acidosis impairs rather than strengthens these functions.

Source: circulation-3:Q28 | Circulation 3, slides 21

24. Which of these locally released substances dilates arterioles?

Answer: A. Histamine

Explanation: Histamine, released from mast cells in injury and allergy, relaxes arteriolar smooth muscle (via H2 receptors) and makes capillaries leakier, which is why inflamed tissue turns red and swollen.

Not B: Endothelin-1 from the endothelium is one of the most powerful constrictors known.

Not C: Serotonin from activated platelets constricts the injured vessel to limit bleeding.

Not D: Angiotensin II constricts arterioles and raises resistance.

Source: circulation-2:Q17 | 6-7

25. An ECG machine runs at 25 mm/s. How much time passes while the paper advances by 1 mm horizontally?

Answer: D. 0.04 s.

Explanation: At a speed of 25 mm each second, 1 mm represents 1/25 s = 0.04 s, the duration of one small horizontal grid square.

Not A: 0.10 s does not correspond to a 1-mm horizontal square at this speed.

Not B: 0.50 s would occupy 12.5 small squares.

Not C: 0.20 s is the time represented by one 5-mm large square.

Source: lecture-7:Q35 | 2, 13

26. A cerebral vessel ruptures and blood enters brain tissue. Which type of cerebrovascular event is present?

Answer: D. Haemorrhagic stroke.

Explanation: Haemorrhagic stroke results from rupture of a vessel, whereas ischaemic stroke results from interrupted arterial supply.

Not A: Hypoglycaemia causes global dysfunction rather than vessel rupture.

Not B: Embolism blocks flow rather than directly representing vessel rupture.

Not C: Thrombosis obstructs an artery instead of rupturing it.

Source: lecture-6:Q18 | 25

27. A ventricle has an end-diastolic volume of 130 mL and an end-systolic volume of 55 mL. What is its stroke volume?

Answer: B. 75 mL.

Explanation: Stroke volume is EDV minus ESV, so 130 - 55 = 75 mL per beat.

Not A: One hundred thirty mL is end-diastolic volume.

Not C: Fifty-five mL is end-systolic volume.

Not D: Adding EDV and ESV does not give stroke volume.

Source: circulation-1:Q22 | 16; Chapter 4 ?7

28. How does CSF substantially reduce the effective weight of the brain?

Answer: B. It provides buoyant support against gravity.

Explanation: Immersion provides buoyancy, reducing the brain's effective supported weight from roughly 1400-1500 g to around 50 g.

Not A: The tissue density does not increase to create buoyancy.

Not C: Brain blood volume remains substantial.

Not D: Lower oxygen consumption does not explain mechanical support.

Source: lecture-6:Q9 | 13-15

29. Why is 1500 used to calculate rate from the number of small squares between R waves at 25 mm/s?

Answer: A. It is the number of 1-mm squares passing in one minute.

Explanation: At 25 mm/s, the paper travels 25 x 60 = 1500 mm per minute. Dividing these 1500 small squares by squares per beat yields beats per minute.

Not B: The distance travelled per cycle varies with RR duration.

Not C: Vertical voltage is calibrated at 10 mm per mV.

Not D: A person's actual heart rate is not fixed at 1500 beats per minute.

Source: lecture-7:Q44 | 2, 13-14

30. A patient with massive haemorrhage remains severely hypotensive after blood volume is completely restored. Which stage fits?

Answer: B. Irreversible shock after extensive tissue injury.

Explanation: Severe myocardial, neural and microvascular damage may prevent recovery even after the original blood-volume loss has been corrected.

Not A: Compensated shock should retain useful mechanisms for pressure support.

Not C: Postural hypotension should depend on standing and impaired compensation.

Not D: An uncomplicated faint generally improves rapidly when lying flat.

Source: circulation-3:Q41 | Circulation 3, slides 25-27

31. A bleeding patient has a nearly normal arterial pressure but cold skin and reduced urine output. What is the best conclusion?

Answer: B. Low perfusion can coexist with normal pressure.

Explanation: Sympathetic arteriolar constriction can maintain arterial pressure despite reduced cardiac output and poor perfusion of some tissues.

Not A: Shock is defined by inadequate tissue perfusion, not one pressure value.

Not C: Haemorrhage reduces venous return and usually lowers output.

Not D: Flow can be redistributed away from skin and kidneys.

Source: circulation-3:Q18 | Circulation 3, slides 12-13; Chapter 20 ?1

32. Why can chronic widespread alveolar hypoxia lead to pulmonary hypertension?

Answer: B. Persistent hypoxic constriction increases lung resistance.

Explanation: When hypoxic constriction affects much of the lung, cumulative resistance rises and pulmonary arterial pressure can increase.

Not A: Recruitment is a different response that lowers resistance.

Not C: The lecture explicitly associates hypoxia with pulmonary vasoconstriction.

Not D: Lung flow generally continues but faces increased resistance.

Source: lecture-6:Q36 | 32

33. Once local metabolic dilation restores an organ's blood flow, what prevents dilation from continuing unchecked?

Answer: C. The restored flow clears accumulated dilator metabolites.

Explanation: The dilators exist because flow was too low for demand. Once flow matches demand, they are washed away as fast as they are made, so the arterioles regain their tone. This negative feedback stops dilation running away.

Not A: Restored delivery generally improves tissue oxygen availability.

Not B: Sustained lactate accumulation would favour continued dilation.

Not D: Vascular smooth muscle remains present and functional.

Source: circulation-2:Q10 | 5-6

34. A graph of organ flow against perfusion pressure is almost flat over a middle range but slopes downward at very low pressures. Why?

Answer: A. The arterioles reach their maximal dilation at low pressure.

Explanation: Once arterioles are fully dilated, they cannot reduce resistance further, so flow decreases as pressure falls.

Not B: Maximal constriction occurs in response to rising pressure, not very low pressure.

Not C: A downward slope does not mean perfusion pressure is zero.

Not D: Venous vessels do not replace arterioles as the chief resistance site.

Source: circulation-2:Q4 | 4-5, 8-11

35. A burn patient presents with shock and an initially high haematocrit. What explains the contrast with recent whole-blood haemorrhage?

Answer: D. Plasma leakage leaves intravascular red cells concentrated.

Explanation: Severe burns can cause plasma to leak through damaged skin while red cells remain, producing haemoconcentration and raised haematocrit.

Not A: Selective red-cell destruction would tend to lower red-cell concentration.

Not B: Plasma loss reduces rather than increases effective circulating volume.

Not C: Erythropoiesis is much too slow to explain this early result.

Source: circulation-3:Q43 | Circulation 3, slides 28

36. The jugular venous 'c' wave is chiefly linked to which event?

Answer: C. Systolic bulging of the tricuspid valve.

Explanation: Early right ventricular contraction can bulge the shut tricuspid valve toward the right atrium, contributing to the c wave.

Not A: Rapid atrial emptying contributes to the y descent.

Not B: Venous filling behind a closed tricuspid valve makes the v wave.

Not D: Right atrial systole generates the a wave.

Source: circulation-1:Q68 | 15, 24-25; Chapter 11 ?4

37. When both atrioventricular valves are open and semilunar valves are shut, which process predominates?

Answer: B. Ventricular filling during diastole.

Explanation: With ventricular pressure below atrial pressure, AV valves open and blood enters relaxed ventricles while semilunar valves remain closed.

Not A: Ejection requires open semilunar valves.

Not C: Outflow valve closure happens near the start of ventricular relaxation.

Not D: Isovolumetric contraction occurs after AV valves close.

Source: circulation-1:Q3 | 11-13

38. Following regional node removal, a patient develops persistent firm limb swelling. What explains it?

Answer: C. Protein accumulation after failed lymph drainage.

Explanation: Lymphatic obstruction impairs removal of interstitial proteins and fluid, leading over time to persistent, often non-pitting swelling.

Not A: Oxygen delivery is not the main mechanism.

Not B: Cardiac ejection fraction need not be abnormal.

Not D: Impaired lymph drainage prevents, rather than accelerates, interstitial protein removal.

Source: circulation-1:Q90 | 94-95

39. A blood-pressure reading is 132/84 mmHg. What is the pulse pressure?

Answer: C. 48 mmHg.

Explanation: Pulse pressure equals systolic minus diastolic: 132 - 84 = 48 mmHg. It measures the pressure swing between systole and diastole.

Not A: Thirty-six is below the arithmetic difference.

Not B: Forty-four is four mmHg below the difference.

Not D: Forty does not equal 132 minus 84.

Source: circulation-1:Q60 | 82

40. Under the Frank-Starling law, why does greater ventricular filling strengthen systolic force?

Answer: B. Stretch of myocardial fibres strengthens contraction.

Explanation: Within the physiological range, more end-diastolic filling stretches fibres and increases developed contraction force and stroke volume.

Not A: Calcium extrusion is related to relaxation, not the primary stretch mechanism.

Not C: Lower venous return reduces ventricular fibre stretch.

Not D: Afterload is a separate determinant of ejection.

Source: circulation-1:Q36 | 38-43